(b) To determine the overall cell reaction, multiply the half-cell equations by factors which balance the electrons in the two equations and add the resulting equations.

$Ag \rightarrow Ag^{+} + e^{-} \quad = \quad x\ 2$

$Br_2 + 2e^{-} \rightarrow 2Br^{-} \quad = \quad x\ 1$

$2Ag \rightarrow 2Ag^{+} + 2e^{-}$

$Br_2^{+} + 2e^{-} \rightarrow 2Br^{-}$

$2Ag \rightarrow 2Ag^{+} + 2e^{-}$

$2Ag + Br_2 \rightarrow 2Ag^{+} + 2Br^{-}$

2. Predict whether the reaction below would occur in aqueous solution in which all concentrations are 1 mol L^{-1}.

$2Fe^{3+} + 2Br^{-} \rightarrow 2Fe^{2+} + Br_2$

(a) Identify half-reactions involved

$Fe^{3+} \rightarrow Fe^{2+}$

$Br^{-} \rightarrow Br_2$

(b) Write half-cell equations and reduction potentials

$Fe^{3+} + e^{-} \rightarrow Fe^{2+} \qquad E° = +0.77\ V$

$2Br^{-} \rightarrow Br_2 + 2e^{-} \qquad E° = -1.09\ V$

(c) Calculate cell voltage

$E° = +0.77 - 1.09$
$= -0.32\ V$

(d) Since E° is negative the reaction will not occur. If a reaction has a positive E° it could occur as written.

SET 19

1. Calculate the standard cell voltages and write the overall chemical reactions for cells which consist of the following half-cells

(a) Zn^{2+} / Zn and Sn^{2+} / Sn

(b) Cr^{3+}/Cr and Ag^{+}/Ag

(c) Hg^{2+}/Hg and Cu^{2+}/Cu

(d) Mg^{2+}/Mg and Cu^{2+}/Cu

(e) Mg^{2+}/Mg and Ag^{+}/Ag

(f) Fe^{3+}/Fe^{2+} and $Cr_2O_7^{2-}/Cr^{3+}$ ($Cr_2O_7^{2-}$ is acidified)

(g) Cl_2/Cl^{-} and I_2/I^{-}

2. Predict whether the following reactions could occur under standard conditions:

(a) $2MnO_4^- + 10I^- + 16H^+ \rightarrow 2Mn^{2+} + 5I_2 + 8H_2O$

(b) $Sn^{4+} + H_2O_2 \rightarrow Sn^{2+} + 2H^+ + O_2$

(c) $Cr_2O_7^{2-} + 6F^- + 14H^+ \rightarrow 2Cr^{3+} + 3F_2 + 7H_2O$

(d) $Cu + 2H^+ \rightarrow Cu^{2+} + H_2$

(e) $2Fe^{3+} + Sn^{2+} \rightarrow 2Fe^{2+} + Sn^{4+}$

3. (a) Which of the following species could react with 1 mol L^{-1} HCl to form hydrogen gas?
   - (i) Cu
   - (ii) Mg
   - (iii) Hg
   - (iv) Ag
   - (v) Sn
   - (vi) Zn

   (b) From the table of reduction potentials supplied, identify
   - (i) a reducing agent which could convert Pb^{2+} to Pb, but not Co^{2+} to Co.
   - (ii) an oxidising agent which could convert Cl^- to Cl_2, but not F^- to F_2.
   - (iii) a reductant which could convert H^+ to H_2, but not H_2O to H_2.
   - (iv) an oxidant which could convert Ag to Ag^+, but not Hg to Hg^{2+}.
   - (v) a reductant which could convert acidified MnO_4^- to Mn^{2+}, but not acidified $Cr_2O_7^{2-}$ to Cr^{3+}.

4. Predict whether the following disproportionation reactions could occur in aqueous solution:

(a) copper(I) ion to copper(II) ion and copper metal
(b) Iron(II) ion to iron(III) ion and iron metal
(c) hydrogen peroxide to water and oxygen gas
(d) chlorine to hypochlorous acid and chloride ion
(e) manganese dioxide to permanganate ion and manganese(II) ion

5. Predict whether reactions could occur in each of the following. Assume standard conditions.

(a) Chlorine gas is bubbled through potassium bromide solution.
(b) Iron(II) nitrate is mixed with sodium iodide.
(c) Aluminium is added to hydrochloric acid.
(d) An iron nail is placed in a tin(II) chloride solution.
(e) An iron(II) sulfate solution is placed in a nickel container.
(f) Hydrogen sulfide is bubbled through an acidified potassium dichromate solution.
(g) Chlorine gas is bubbled through an acidified solution of barium nitrate.
(h) Chlorine gas is bubbled through an acidified solution of iron(II) bromide.

30. $NH_4NO_2 \rightarrow N_2 + 2H_2O$

31. $3I_2 + 6OH^- \rightarrow IO_3^- + 5I^- + 3H_2O$

32. $4MnO_4^- + 4OH^- \rightarrow 4MnO_4^{2-} + O_2 + 2H_2O$

Set 19

1. (a) $Zn + Sn^{2+} \rightarrow Zn^{2+} + Sn$, +0.62V
   (b) $3Ag^+ + Cr \rightarrow 3Ag + Cr^{3+}$, +1.54V
   (c) $Hg^{2+} + Cu \rightarrow Hg + Cu^{2+}$, +0.51V
   (d) $Mg + Cu^{2+} \rightarrow Mg^{2+} + Cu$, +2.70V
   (e) $Mg + 2Ag^+ \rightarrow Mg^{2+} + 2Ag$, +3.16V
   (f) $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 7H_2O + 6Fe^{3+}$, +0.56V
   (g) $Cl_2 + 2I^- \rightarrow 2Cl^- + I_2$, +0.82V

2. (a) Yes, +0.99V
   (b) No, - 0.53V
   (c) No, -1.54V
   (d) No, -0.34V
   (e) Yes, +0.62V

3. (a) Mg, Sn, Zn
   (b) (i) Sn or Ni
   (ii) H_2O_2 or MnO_4^- (these are common oxidants, both must be acidified)
   (iii) Pb, Sn, Ni, Co
   (iv) $O_2/4H^+$
   (v) Au, Cl^-/H_2O, Cl^-

4. (a) Yes, +0.37V
   (b) No, -1.21V
   (c) Yes, +1.10V
   (d) No, -0.27V
   (e) Yes, +0.63V

5. (a) $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$, +0.27V
   (b) No reaction
   (c) $2Al + 6H^+ \rightarrow 2Al^{3+} + 3H_2$, +1.66V
   (d) $Fe + Sn^{2+} \rightarrow Fe^{2+} + Sn$, +0.30V
   (e) No reaction
   (f) $Cr_2O_7^{2-} + 3H_2S + 8H^+ \rightarrow 2Cr^{3+} + 3S + 7H_2O$, +1.19V
   (g) No reaction
   (h) Both Br^- and Fe^{2+} are oxidised

Set 20

		Anode	Cathode	Reaction	Minimum Voltage
1.	(a)	I_2	Cu	$Cu^{2+} + 2I^- \rightarrow Cu + I_2$	0.20V
	(b)	Br_2	H_2	$2H^+ + 2Br^- \rightarrow H_2 + Br_2$	1.09V